Ayush Singh Parihar

ayushparihar560@gmail.com | +91 8932001247 | linkedin.com/in/ayush-parihar2005 | github.com/AyushSingh2024

Professional Summary

B.Tech Computer Science (AI-ML) undergraduate with strong foundations in Python, Java, and full-stack development. Built agentic AI systems, RAG pipelines, and scalable REST APIs across 3+ end-to-end projects, achieving up to 88% model accuracy and shipping production-style backend systems with secure authentication.

Education

B.Tech in Computer Science & Engineering (AI-ML), UPES, Dehradun **Jul 2023 – Jun 2027**
CGPA: 7.38/10

12th (CBSE), Mahatma Hansraj Modern School, Jhansi **Apr 2022 – Jun 2023**
73.4%

Technical Skills

Languages:	C, C++, Java, Python, JavaScript, TypeScript
AI/Agentic:	LLM APIs (OpenAI, Claude, Gemini), RAG pipelines, vector & vectorless retrieval, agentic tool-calling, MCP, prompt engineering
Web/Backend:	Node.js, Express.js, FastAPI, MERN Stack, MongoDB
Data Science/ML:	scikit-learn, XGBoost, pandas, matplotlib, seaborn, PyTorch (CNNs)
DBs, APIs & Tools:	Qdrant, MySQL, JDBC, Git, Prisma, Razorpay, Docker

Internships

Software Development Intern, Universia, Dehradun **Feb 2025 – Mar 2025**

- Processed and cleaned 15+ satellite and astronomical datasets using Python, reducing preprocessing pipeline runtime by ˜30% through vectorized pandas operations.
- Assisted in real-time telescope operations and data collection across 10+ observation sessions, improving data reliability and reducing measurement error rates for downstream analysis.
- Led and mentored 5+ stargazing sessions for student cohorts of 20+, strengthening technical communication and stakeholder engagement skills.

Projects

VitalRAG – Agentic Health-Document RAG Copilot
Python, FastAPI, Qdrant, LLM APIs (OpenAI/Claude), Mem0 *github.com/AyushSingh2024/vitalrag*

- Designed a hybrid retrieval pipeline combining vector search (Qdrant, semantic chunking, metadata filtering, reranking) with an LLM-generated document summary index to fix multi-document reasoning failures that pure vector RAG missed.
- Engineered a tool-calling agent that routes queries across vector search, summary-index search, and persistent memory, with confidence guardrails that request clarification when evidence is insufficient.
- Implemented a cross-session memory layer (Mem0) so the agent retains user-specific context instead of resetting state per session, and benchmarked retrieval accuracy, latency, and per-query cost across 20+ test queries.

Intelligent Transportation System (ITS)
Python, FastAPI, pandas, NumPy, SUMO, JavaScript *github.com/AyushSingh2024/ACO-for-ITS*

- Developed a high-performance REST API with FastAPI and pandas to process large-scale vehicle trajectory data and visualize real-time transit simulations.
- Implemented Ant Colony Optimization (ACO) algorithms to dynamically compute low-congestion vehicle routes, outperforming Dijkstra's algorithm by up to 22% in congestion-heavy simulations via pheromone-based feedback loops.
- Designed a Particle Swarm Optimization (PSO) system that autonomously adjusts traffic signal timings, achieving 88% accuracy in minimizing intersection wait times across test simulations.

Course Study Assistant - Advanced Video-Transcript RAG
Node.js, LangGraph, Qdrant, PostgreSQL, OpenAI, Cohere **GitHub**

- Created a LangGraph-orchestrated study assistant for 87 React Native and Expo lessons across 18 modules, with timestamp-aware ingestion across subtitle, PDF, JSON, and Markdown sources.
- Implemented hybrid retrieval with Qdrant vector search and PostgreSQL full-text search, combining query expansion, yDE, Reciprocal Rank Fusion, and Cohere reranking.
- Added lesson-order spoiler filtering, future-content lookahead, timestamped citations, and an LLM grader that retries answers with weak grounding or inaccurate citations.